

Data audit: pair formation for synthetic KEP pool

100 simulations

September 8, 2026

Setup

We generate 100 simulated pools of incompatible donor–recipient pairs. Each simulation:

- Samples one HLA profile per person from the HaploStats imputation distribution. If the person has fully observed HLA, the real profile is used (no sampling).
- Evaluates ABO and DSA incompatibility under the sampled profiles.
- Re-indexes donors and patients (random permutation) before building the bipartite graph of incompatible pairs and solving a maximum bipartite matching.

Source dataset: 1,736 patients and 1,179 donors. Each simulation uses a different random seed for HLA sampling. In this version, donors and patients are re-indexed via a random permutation before constructing the bipartite incompatibility graph.

1 Pool size and why

The pool size is 1,087 across the simulations, with 92 unmatched donors (all type O). This count follows from the structure of the bipartite incompatibility graph and the cardinalities of the donor and patient subsets (740 O donors, 759 antibody bearing patients, 439 non-O donors) which are fixed under HLA sampling.

Type O donors are universal donors, so they are never ABO incompatible and can only enter the pool via DSA. The 439 non-O donors are matched because each has many ABO-incompatible patient options. The 759 antibody-bearing patients are shared between two groups: a fraction is absorbed into ABO only matches with non-O donors, and the rest is used by O donors via DSA. After this competition, 648 O donors are matched and 92 are not.

HLA sampling perturbs the identity of individual edges but does not change the cardinalities of these subsets. Since the maximum matching of a bipartite graph is governed by these cardinalities, the size of the pool is constant across simulations. The 92 unmatched donors are not the same in every simulation. Across the 100 runs, 454 donors are matched in every simulation, 725 donors are excluded between 1 and 49 simulations, and no donor is excluded in 50 or more simulations. The identity of the unmatched donors rotates according to the sampled HLA profiles and the re-indexing, while the count remains fixed. See table 7 too.

2 Person rotation across simulations

Table 1: How often each person appears across the 100 simulations.

	Patients		Donors
Appear in ≥ 1 pool	1,687 / 1,736 (97.2%)	1,179 / 1,179 (100%)	
Appear in all 100 sims	639		454
Appear in only 1 sim	0		0
Median appearances (out of 100)	60		91

Most people rotate in and out of the pools. The patients in all 100 sims are highly sensitized or ABO-restricted recipients, the donors in all 100 are the ones with the broadest set of incompatibility edges.

3 HLA profile variation per person

Table 2: Number of distinct HLA profiles sampled per person across appearances.

	N	Median distinct profiles sampled	Max
Patients with one available profile	1,445	1	1
Patients with > 1 imputations	242	2	12
Donors with one available profile	2	1	1
Donors with > 1 imputations	1,177	5	41

We need to take into account that most of the patients (83%) have fully observed HLA or just one possible imputation, so their profile is fixed across simulations. The main source of HLA variability is the donor side: 1,177 donors have multiple imputations and a typical donor is sampled across 5 distinct profiles.

4 Pair level repetition

Table 3: Repetition of pairs across the 100 simulations.

Total pair appearances	108,700
Unique pairs	87,102
Pairs appearing in exactly 1 sim	74,038 (85.0%)
Pairs appearing in 2–5 sims	12,420 (14.3%)
Pairs appearing in > 5 sims	644 (0.7%)
Max times a single pair appeared	60
Repeated pairs with same HLA combination in every appearance	7,178 (54.9%)
Repeated pairs with different HLA combinations	5,886 (45.1%)

The mean Jaccard index between sim 0 and the other 99 simulations is 0.005, so the pools are essentially disjoint at the pair level. Among the 15% of pairs that do repeat, about 45% appear with different HLA combinations.

5 ABO distributions

Table 4: ABO blood type distribution in the formed pairs vs in the available dataset (percentages).

ABO	Patients (%)		Donors (%)	
	in pool	available	in pool	available
O	49.2	41.2	59.6	62.8
A	31.9	40.6	30.7	28.3
B	16.3	13.8	7.6	7.0
AB	2.6	4.5	2.0	1.9

Patients with blood type O are over represented because they are hardest to match. The opposite for the donors.

Table 5: ABO absolute counts: in the available dataset vs average per pool (across 100 simulations).

ABO	Patients		Donors	
	in dataset	per pool (avg)	in dataset	per pool (avg)
O	715	535	740	648
A	704	347	334	334
B	239	177	83	83
AB	78	29	22	22
Total	1,736	1,087	1,179	1,087

Table 6: Donor ABO and patient ABO over the 100 simulations. Counts of pair appearances.

Donor ↓	Patient →	A	AB	B	O	Total
A		75	22	8,807	24,496	33,400
AB		2,187	0	5	8	2,200
B		8,262	0	0	38	8,300
O		24,132	2,829	8,910	28,929	64,800
Total		34,656	2,851	17,722	53,471	108,700

Table 7: Pair appearances by donor ABO, patient ABO and incompatibility reason.

Donor	Reason = ABO (43,803 pairs)			Reason = DSA (64,897 pairs)			
	Pat A	Pat B	Pat O	Pat A	Pat AB	Pat B	Pat O
A	0	8,807	24,496	75	22	0	0
AB	2,187	5	8	0	0	0	0
B	8,262	0	38	0	0	0	0
O	0	0	0	24,132	2,829	8,910	28,929

6 CPRA distribution and highly sensitized patients

Table 8: CPRA distribution: patients in pool vs available.

CPRA bin	in pool (%)	available (%)
0	40.7	58.4
1–20	16.4	13.0
21–50	12.5	8.9
51–80	10.1	6.7
81–99	14.2	9.1
100	6.1	3.9

Table 9: Highly sensitized patients (CPRA > 80) in the simulations.

Highly sensitized patients in dataset	216
Appear in ≥ 1 simulation pool	216 (100%)
Appear in all 100 sims	214

Patients with CPRA = 0 are under-represented in the pool (no antibodies, so they can only be incompatible via ABO). Every highly sensitized patient enters at least one pool, and 214 out of 216 are present in all 100 simulations.

7 Ethnicity distribution: pool vs available

Table 10: Ethnicity distribution in pool vs available patients.

	Patients (%)	
	in pool	available
Caucasian	56.5	57.9
Afroamerican	18.9	17.2
Latin	18.1	17.8
Asian	4.5	5.1
Am. Indian	1.8	1.8
Pac. Islander	0.2	0.2

8 Antibody load per patient

Table 11: Antibody load (UNACC entries per patient).

	Patients in pool	All patients
With at least one antibody	751 / 1,687 (44.5%)	759 / 1,736 (43.7%)
Median antibodies among patients with any	5	5
Max antibodies (single patient)	78	78

9 Demographics of matched people

Table 12: Demographic summary across all pair appearances.

	Patients	Donors
Median age	53	43
Mean age	51.3	43.4
Age IQR	[42, 63]	[34, 53]
Gender	57.4% M, 42.6% F	33.2% M, 66.8% F
On dialysis	74.7% yes, 25.3% no	–